A finite configuration-free partial system has non-negative predimension

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Editorial note. The counted statement is the author's environment at source0.tex lines 402-402 and its proof environment opens at line 404; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; identity is the author label, or the pinned source position where the author gave none.

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Editorial scope. One original result of the pinned author manuscript, counted once: non-negativity of the predimension of a finite configuration-free partial system (unlabelled lemma at source0.tex line 402).

Allowed standard prerequisites (editorial).

- Partial Steiner triple system. A finite structure with a partial operation whose defined products are the blocks.. It is the structure whose predimension is computed.
- Unconfined configuration. A finite configuration is confined when it forces a product that is not already present; containing none is the hypothesis.. It is what allows an HF-ordering to exist.

- Predimension. The predimension delta of a finite system is the number of elements minus the number of blocks.. It is the quantity shown to be monotone along the ordering.

Lemma 0.1 The predimension of a finite partial STS A that contains no confined configuration is non-negative.

Proof. Suppose $|A| = n$. Since A is unconfined, we can find an HF-ordering

$$a_1 < a_2 < \ldots < a_n \,.$$

Let $A_i := \{a_1, \ldots, a_i\}$ for $1 \leq i \leq n$. Then, if $|\mathrm{b}(A_i)|$ is the number of blocks in A_i,

$$\delta(A_i) = \begin{cases} |A_{i-1}| - |\mathrm{b}(A_{i-1})| + 1 - 1 = \delta(A_{i-1}) & \text{if } a_i \text{ belongs to one block in } A_i \\ |A_{i-1}| - |\mathrm{b}(A_{i-1})| + 1 & \text{if } a_i \text{ belongs to no blocks in } A_i \,. \end{cases}$$

Therefore $\delta(A_i) \geq \delta(A_{i-1})$ for all i. □